

04.13.26 | week 3 | SWE 263P

UX Research & Deliverables

Dr. Matthew Bietz

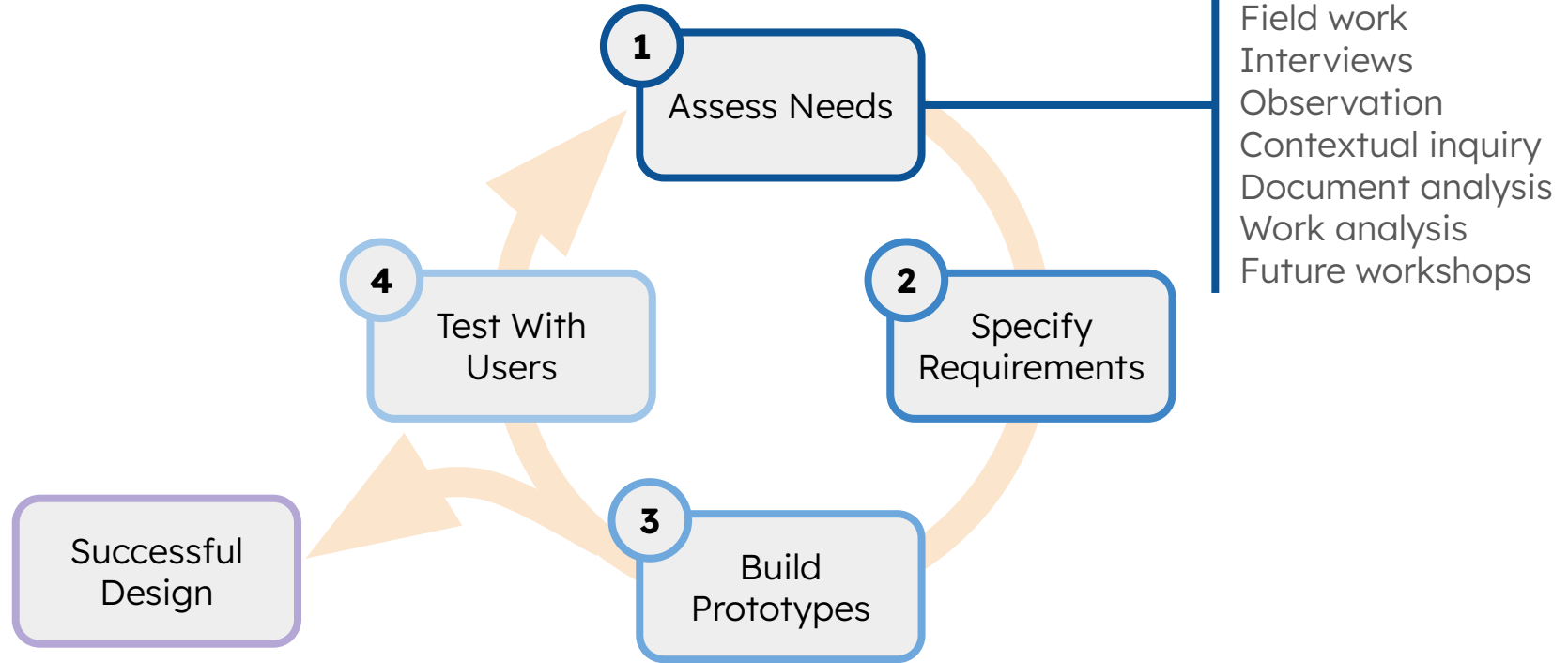
UCI Department of Informatics
Donald Bren School of Information & Computer Sciences

Includes content developed by Prof. Stacy Branham

Path

- UX Research (~20min)
- Interviews, Observations, & Contextual Inquiry (~45min)
- Activity 1 + Break (~15min + 15min)
- Personas & Journey Maps + Activity 2 (~15min + 50min)
- Coming Up (~5min)

Phase 1: Assess



Needfinding via Ethnography



**What lessons
do you take
from this
video?**





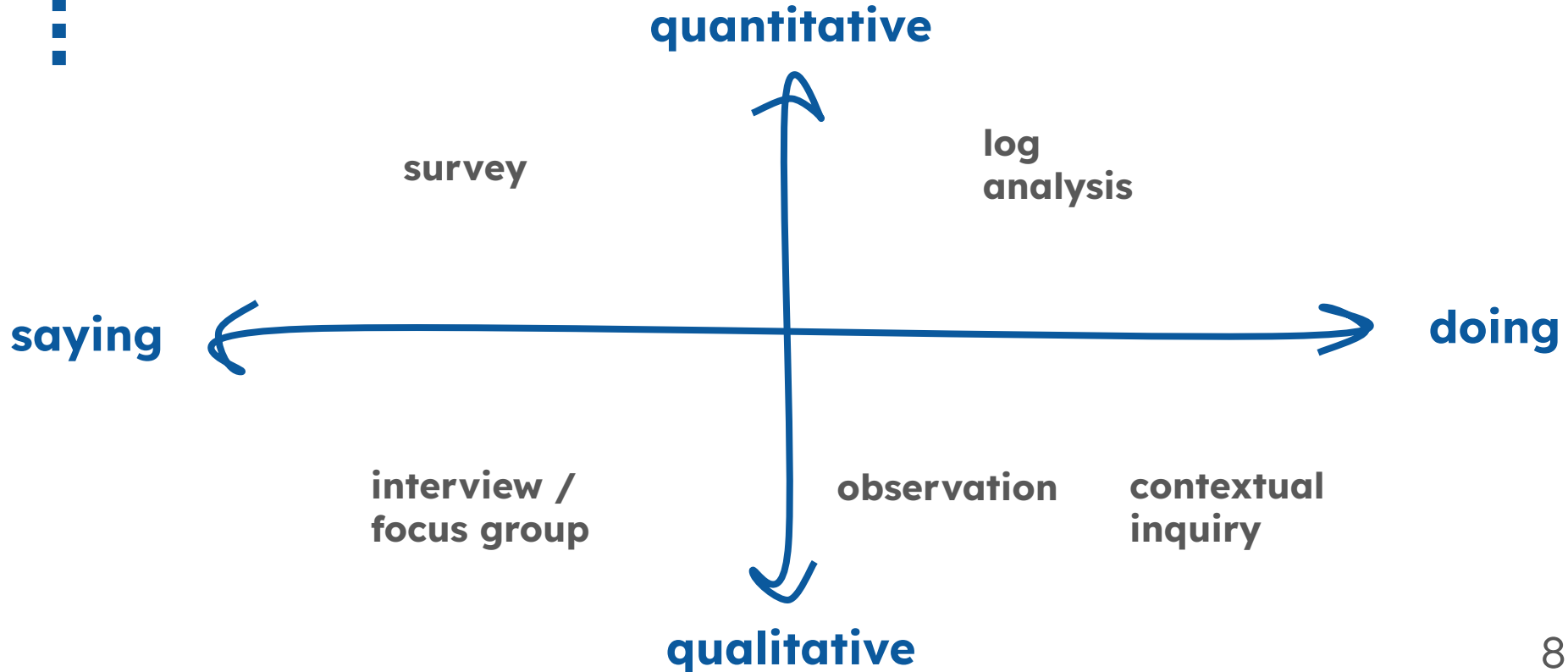
What did you notice?

- › Parking in cities ->
 - Current signs show restrictions instead of what's allowable → “can’t figure out if I can park here”
 - Loading zones are inefficient → “there’s not enough parking”
 - New signs used color and icons more effectively → is there a cost constraint?
 - Dynamic parking meters
- › Observing people’s behavior:
 - Naive state of mind
 - Chaos
 - Time, patience
 - “Hidden obvious”
- › “What people say, what people do, and what people say they do are entirely different things.”

Needfinding & User Research

- › People are **notoriously bad** at predicting what they would use, or would prefer when it is only hypothetical
- › They can respond much better *in situ* to actual, **concrete** things, or make comparisons
- › This highlights the importance of **contextual interviews**, **observation**, and **prototypes**

(Some) Needfinding Approaches



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Interviews vs. Observations?

> Similarities

- Interviews and observations require **interpretation**
- You can elicit the information you are most interested in through follow-up

> Differences

- Interviews are typically 1-on-1; observation can be 1-to-many
- Interview gathering verbal / textual information → observation audio and textual information / field notes / pictures / maps



Interviews

- › **Who?** 1 or more interviewers, 1-2 interviewees
- › **What?** various levels of question preparation
- › **When?** Between 5-120 minutes; negotiated
- › **Where?** In-person; remote phone/video chat; epistolary (e.g., via email); in situ; In a “third space”?



Interviews: Formats

› Interviews can be:

- structured: pre-planned questions, read verbatim, not read out of order
- semi-structured: prepared questions, but the interviewer expects to deviate from them, asking follow-up questions and going out of order, so as to follow the interviewee's lead
- unstructured: spontaneous questions; may have a topic chosen, but conversation is open and free-flowing

› Interview questions can be:

- open: answers are more likely to be longer and diverse between users (e.g., “Tell me about your career path.” or “What’s your favorite part of your job?”)
- closed: possible answers are chosen from a given or implied discrete list (e.g., “what’s your job title?” or “do you own a mobile device?”)



Interviews: Process

- › Develop a protocol
 - materials (e.g., notebook, pencil, recording devices x 2)
 - reminders (e.g., turn on the recorders, gain consent)
 - questions, timing info
- › Run a “pilot” or test interview(s)
- › Recruit participants
- › Negotiate location, communication tools
- › Gain user consent (e.g., adult) / assent (e.g., child)?
 - privacy
 - safety
 - compensation



Interviews: Pros/Cons of Interviews

- › detailed qualitative data
- › elicitation allows for follow-up on interesting leads
- › first-person accounts
 - gives some users a say, but may misrepresent the larger population
 - can be more OR less trustworthy, depending on the type of information
- › take a lot of time, but relatively less than an observation
- › requires a lot of time to analyze

Interview Demo

Technology + Food



Demo: What Did You Notice?

- › Open-ended question... to start...
 - Get people comfortable with talking to you
 - See the emotions / values
- › Follow up questions
 - Unstructured interview... all my questions were follow-ups
- › Time constraint - be monitoring the time / respectful
- › Share a bit about myself (interviewer) in the dialogue
 - Building rapport (build a relationship)
 - Cut down on your time, can make it more difficult to connect / bias
- › Set up the next interview
- › “May I start recording now”
- › Take notes



Ethnographic Observations

- › **Who?** Researcher embedded in a community
- › **What?** Observe; take field notes; gather documents; interpret; participate
- › **When?** From months to years!
- › **Where?** Wherever the users are—as localized as a specific room to as diffuse as an online community

Ethnographic Observation Process

- › Need to gain access to a field “site”; requires **buy-in** from community
- › Develop “**detached partnership**” with community
- › Take extensive “**field notes**”
 - Includes both direct observations as well as direct interpretations.
 - May combine multiple data formats: timestamps, written descriptions, diagrams, photographs, audio/video, etc.
 - Expand field notes within ~24 hours
- › Conduct thematic analysis, write-up





Fieldnotes From My Work - 1

One bearded man, probably in his late 50s or early 60s, shows off the little clip that he's wearing on his shirt. He tells us that it's a new device, funded through a kickstarter campaign, that takes a picture every 30 seconds to create a photo stream for your day. He just got his device today, and is trying it out for the first time. He's a doctor, and he's trying it out as a possible clinical intervention. "I find if a technology works for me, it will usually work for my patients." A newcomer joins the group, wearing the same device as the doctor. They recognize the clip on each other, and have a moment of geek recognition. They immediately launch into conversation about the device - when did you get it, has it been useful, what are its quirks, etc..



Fieldnotes From My Work - 2

[He] tells us about the work he'd done with the [year] Women's Olympic [sport] team to use off-the-shelf, low-budget tracking devices to help them win a silver medal. Throughout his presentation he refers to the team as "the girls." His wife stands silently by his side as he talks. She never speaks.



Pros / Cons of Observation?

- › *Extremely* detailed qualitative data
- › Elicitation allows for follow-up on interesting leads
- › First-person accounts *plus* direct observation
 - Gives researcher access to what users can't or won't say
 - Can improve reliability of interpretations
- › Can take a lot more time than interviewing to both gather and analyze data



Contextual Inquiry (CI)

“The core premise of contextual inquiry is very simple: go where the customer works, **observe** the customer as he or she works, and **talk** to the customer about the work. Do that, and you can’t help but gain a better understanding of your customer.”
(Beyer & Holtzblatt, 1998)

CI is, effectively, **observation + interview**



Human Relationships

- › It can be easy for user researchers slip into counterproductive relationships with the end user
 - interviewer / interviewee relationship
 - guest / host relationship
 - expert (you) / novice (them)

Goal: Master-Apprentice Relationship

the “user” is the master, the designer the apprentice





Four Principles of CI

› **Context**

“go where the work is” & stay “concrete”

› **Partnership**

“help customers articulate their work experience”

› **Interpretation**

“determine what customers words & actions mean together”

› **Focus**

“clear focus steers the conversation” but also “conceals”

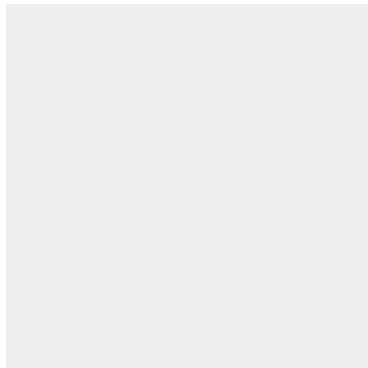


CI Process

- › Start with a **conventional interview** (~15 min)
- › **Transition** to active observation (~15 seconds)
- › The **contextual interview** proper (~2 hours)
- › **Wrap up** by checking your interpretations (~15 min)

Path

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- Coming Up (~5min)



[Activity 3.1](#)

 15 min

In-Class Activity 1

User Research Methods

- › Pair up (exactly 2 people/group)
- › Each chooses a different topic and IDEO method card and prepares a short research plan.
- › Conduct the research, take notes, and be ready to use them in the next exercise.

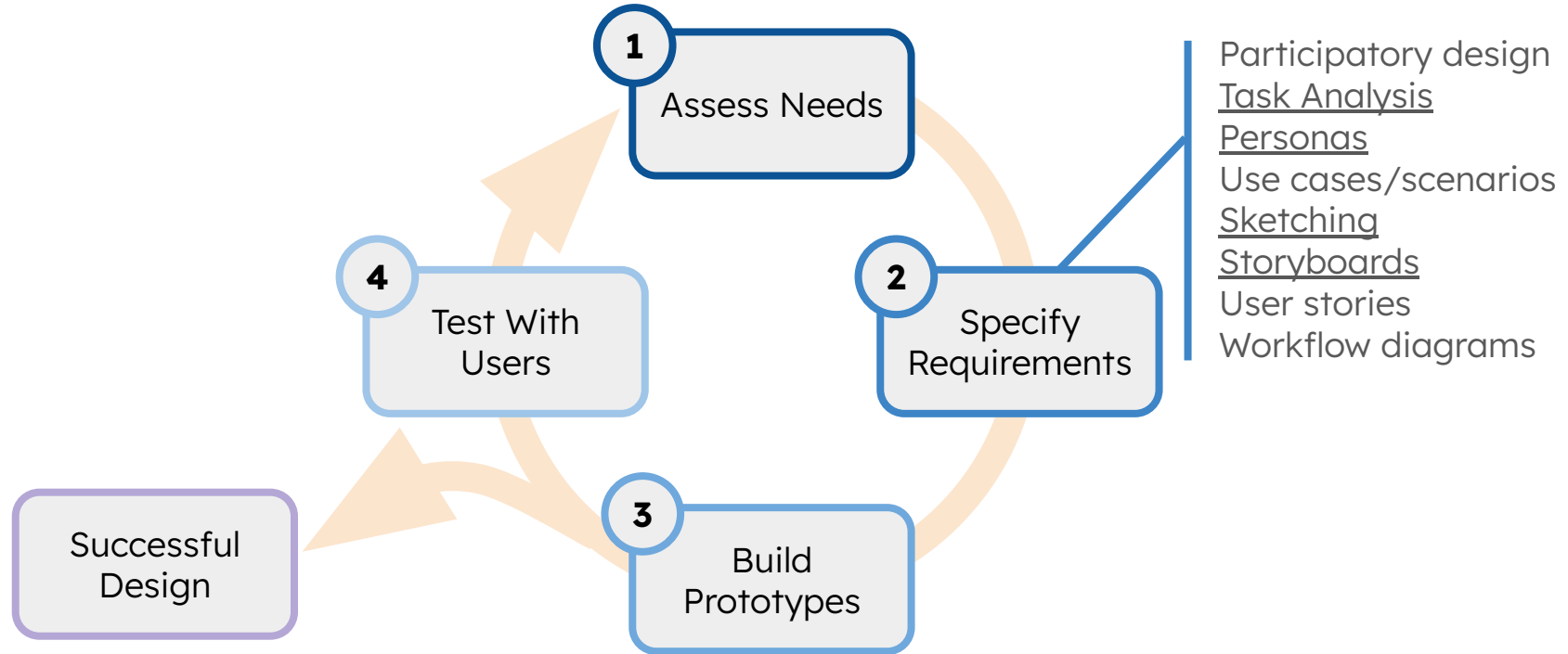
15 min break

Come back at 5:45

Path

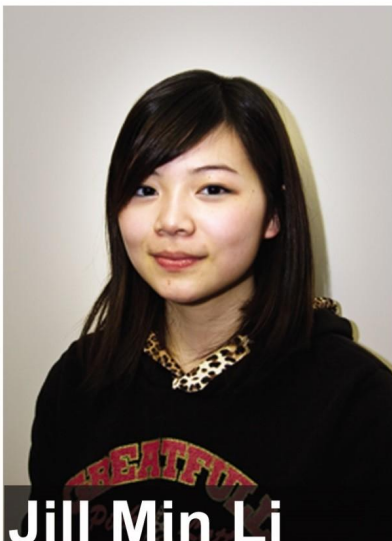
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- Coming Up (~5min)

Phase 2: Specify



PERSONA

User Group -
Male & Female aged 18 - 25 international Students studying in UK



Jill Min Li

Age: 20
Gender: Female
Marital Status: Single
Ethnic Origin: Korean Chinese Mix
Occupation: University student
Year 2 Undergraduate
Place: Private Residential Apartment
Interests: Cooking, Socialising, TV & Film,
Leisure Sport, Shopping & Travelling.

Lifestyle

Jill has been studying in UK for 7 years and has been very dedicated in her education as an overseas student. The years away, she has been socialising with her friends. At first her cooking skills were basic but gradually improving. In which at this moment, has helped her introduce her personality, through the food she cooks and shares with friends.



Quotes

"I hate eating alone"

"To me, cooking is a great way to meet new friends"

"It is very hard to find ingredients that are from home,
so I will get a lot at a time but never finish them all."

"When I am cooking for myself
I will cook quickly but try to stay healthy"

"I prefer my kitchen to be clean at all times
but I never do the washing up after cooking"

GOALS

- Effectively using "Home Food" to produce well cooked meals
- Intergrating own food culture and experience to share with her friends.
- Socialise and creates great memory from the gathering
- Being more organised and avoid building mess after cooking.
- To create exciting and presentable dishes.
- Eating healthy



Name:	TJ Thyne
Occupation:	Co-founder and Chief of Marketing
Company Size:	Startup (Team of 5)
Education:	MA in Communication
Age:	31

Goals

1. Spreading the message about his new startup,
2. monitoring the web for any mention about his new startup and competitors' actions
3. providing great tutorials
4. increasing new sign ups

Needs

1. Fast and easy tool to manage Social Media
2. Mobile device support and app
3. Collaborative tool
4. Metrics on the go

Technical background and workplace

TJ lives and breathes the Web. It's easier to mention the websites he doesn't visit every day. Thinking about it, he's interested in everything from growth hacking to copywriting. He's a one man marketing machine for his startup. He works from his home office, Starbucks and even from a beer joint. He's permanently connected and working. He loves to travel and runs to relax. He is active on Twitter and Quora, a Facebook addict and Instagram junkie followed by over three thousand people. He uses iPad mini, iPhone 5 and MacBook Air (retina). He loves products that are usable and well designed.

TIMO THE SERVICE-MINDED OPTIMIST



MALE

29-31 YEARS OLD

LIKES FOREIGN CARS

SOME COLLEGE EXP.

MAKES \$19K - \$24K PER YEAR

Timo is a hard-working guy in his late twenties, living in the North Hill area of Akron. He's got a job he likes and feels that if he continues to work hard, his life will continue to get better. He lives by the mantra that today is better than yesterday and, as long as he stays focused, tomorrow will be better than today. He's lived through some tough times and as the result of some bad financial decisions, did some pretty serious damage to his credit rating. Timo has a general understanding of how credit works but he didn't realize how much it would affect him until he needed to make a purchase that required him to get a loan.

He is very protective of his long-time girlfriend Eryn so he wants to provide her with the safety and reliability of a newer

"If I'm dropping a ton of money on something like a car, I expect top service and to be treated with respect. Absolutely without question. If people are spending a lot of money, service needs to be there."

GOALS & MOTIVATIONS

Find the absolute best car for lifestyle & budget

Buy something reliable (not driven by price)

Feel confident in the decision to purchase

Be treated with respect

Rebuild his credit

Sophie Behar

Business Professional

"I am better at my job because of my BlackBerry, but I will NEVER be a "CrackBerry" addict!"

Age Demographic

28-31

Life Stage

Career builder; married, one

Internet Usage

kid 5 y/o

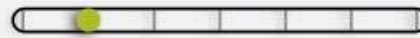
Leisure Reading

10hrs per week

Relevant Products

Management books
BlackBerry device;
Mustung; Roxio; BIS

Extrovert



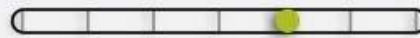
Introvert

Sensing



Intuition

Thinking



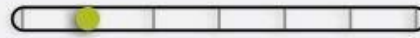
Feeling

Judging



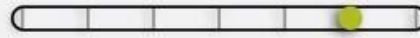
Perceiving

Trendy



Status Quo

Sub Culture



Main Stream

Liberal



Conservative



Personas Generally Include...

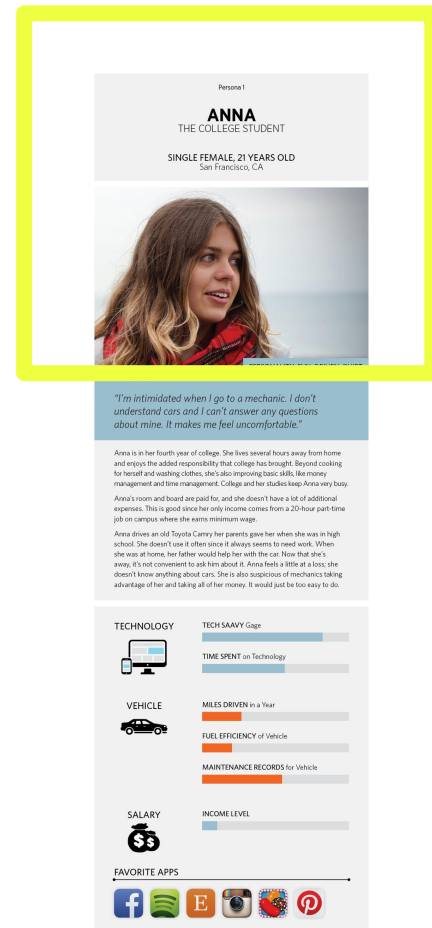
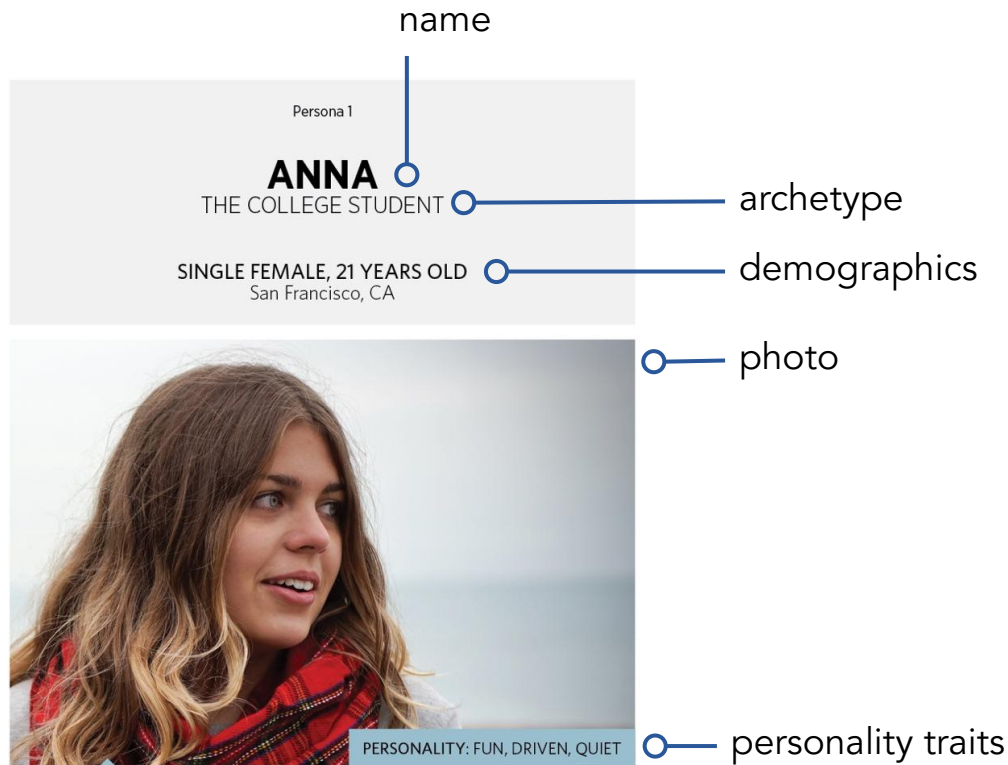
- › Background demographics (age, income, ethnicity, gender)
- › Photo
- › Personality
- › Goals
- › Quote → real
- › Technologies used & pain points
- › Archetype
- › Background paragraph



What's a Persona?

- › **Fictitious** representations of users
- › Based on user **research** insights
- › Tools for designers / engineers
- › **Facilitate discussions** over what features to include for what user groups

Personas include...



Personas include...

"I'm intimidated when I go to a mechanic. I don't understand cars and I can't answer any questions about mine. It makes me feel uncomfortable."

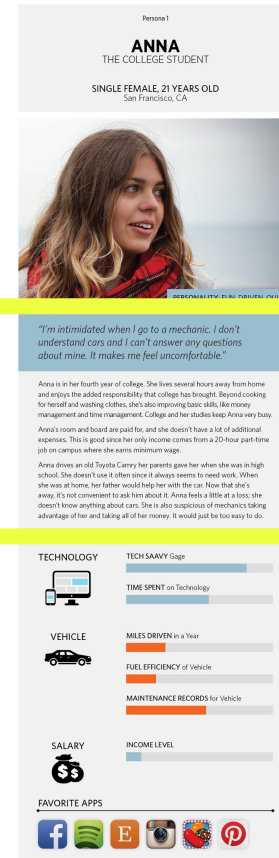
Anna is in her fourth year of college. She lives several hours away from home and enjoys the added responsibility that college has brought. Beyond cooking for herself and washing clothes, she's also improving basic skills, like money management and time management. College and her studies keep Anna very busy.

Anna's room and board are paid for, and she doesn't have a lot of additional expenses. This is good since her only income comes from a 20-hour part-time job on campus where she earns minimum wage.

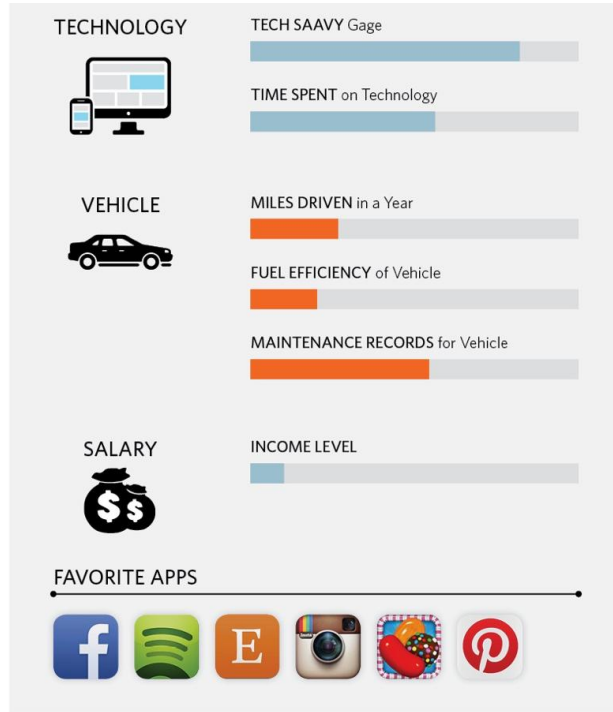
Anna drives an old Toyota Camry her parents gave her when she was in high school. She doesn't use it often since it always seems to need work. When she was at home, her father would help her with the car. Now that she's away, it's not convenient to ask him about it. Anna feels a little at a loss; she doesn't know anything about cars. She is also suspicious of mechanics taking advantage of her and taking all of her money. It would just be too easy to do.

— revealing quote

— narrative



Personas include...



Personal traits / skills

behavior, etc.

tech use

Persona 1

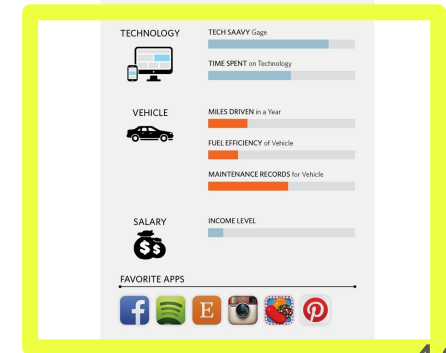
ANNA
THE COLLEGE STUDENT

SINGLE FEMALE, 21 YEARS OLD
San Francisco, CA

PERSONALITY: FUN, DRIVEN, QUIET

"I'm intimidated when I go to a mechanic. I don't understand cars and I can't answer any questions about mine. It makes me feel uncomfortable."

Anna is in her fourth year of college. She lives several hours away from home and enjoys the added responsibility that college has brought. Beyond cooking for herself and washing clothes, she's also improving basic skills like money management and time management. College and her studies keep Anna very busy. Anna's room and board are paid for, and she doesn't have a lot of additional expenses. This is good since her only income comes from a 20-hour part-time job on campus where she earns minimum wage. Anna drives an old Toyota Camry her parents gave her when she was in high school. She doesn't use it often since it always seems to need work. When she was at home, her father would help her with the car. Now that she's away, it's not convenient to ask him about it. Anna feels a little at a loss; she doesn't know anything about cars. She is also suspicious of mechanics taking advantage of her and taking all of her money. It would just be too easy to do.



Personas come in sets

THE USERS



Example from accessibility research



Rachel Lee
Age: 64
Work: Systems Advocate
Family: Married
Location: Marietta, GA

Preferences

- H** Technological Assistance vs Human Assistance
- P** Planned Routes vs Spontaneous Routes
- F** Familiar Technology vs New Technology
- En** Enjoyable Experience vs Efficient Experience

Bio

Rachel works in a Systems Advocacy department where she works to make policies fair for all people. Throughout the week, she mostly goes out for work, but will occasionally drive out to shop, eat, or go to a bar. Rachel was born with spinal bifida, but utilizes her power wheelchair and modified car to be able to navigate.

Transportation



- Walking
- Car
- Taxi/Ride Share
- Paratransit
- Bus/Metro
- Solo
- With Companion

Assistive Aids



Modified Car
Power Wheelchair
Google Maps

Goals

- Wants to be able to easily navigate in unfamiliar locations and terrain
- Requires planned routes to be completely accessible for wheelchairs
- Would like to navigate with minimal assistance

Desires

- Maps that provide routes based on mobility needs
- Maps that indicate accessible features such as curb cuts, elevators, and accessible parking
- Ability to easily see how well a road is paved
- A way to know how crowded a path or destination will be

Pain Points

- Challenging to find accessible features
- Avoids steep ramps and large crowds
- Cannot use power wheelchair in unfamiliar terrain
- Difficult to find accessible features based on directions from other people

Omar
Age: 60
Occupation: Policy Consultant
Family: Married
Location: Arlington, VA

Bio

Omar is a policy consultant who works for the federal government. He is a frequent traveler and often has to navigate unfamiliar terrain. He is a frequent traveler and often has to navigate unfamiliar terrain. He is a frequent traveler and often has to navigate unfamiliar terrain.

Goals

- Wants to be able to easily navigate in unfamiliar locations and terrain
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Pain Points

- Challenging to find accessible features
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- Cannot use power wheelchair in unfamiliar terrain
- Difficult to find accessible features based on directions from other people

Brandon
Age: 40
Occupation: Systems Administrator
Family: Single
Location: Houston, TX

Bio

Brandon is a systems administrator who works for a large corporation. He is a frequent traveler and often has to navigate unfamiliar terrain. He is a frequent traveler and often has to navigate unfamiliar terrain. He is a frequent traveler and often has to navigate unfamiliar terrain.

Goals

- Wants to be able to easily navigate in unfamiliar locations and terrain
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Desires

- Maps that provide routes based on mobility needs
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Pain Points

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Alice Hughes
Age: 70
Occupation: Retired
Family: Single
Location: San Antonio, TX

Bio

Alice is a retired teacher who lives in a small town. She is a frequent traveler and often has to navigate unfamiliar terrain. She is a frequent traveler and often has to navigate unfamiliar terrain. She is a frequent traveler and often has to navigate unfamiliar terrain.

Goals

- Wants to be able to easily navigate in unfamiliar locations and terrain
- Requires planned routes to be completely accessible for wheelchairs
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Desires

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Pain Points

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Persona content matched to purpose

Preferences

- H** **Human Assistance**
preferred over Technological Assistance
- P** **Planned Routes**
preferred over Spontaneous Routes
- F** **Familiar Technology**
preferred over Latest Technology
- En** **Enjoyable Experience**
preferred over Efficient Experience

- › Content of personas should be both relevant to your purpose, but also broad enough to represent a holistic person



Why create personas?

- › “The user is not like me”
- › Helps designers imagine how their tool will be used by others
- › Bridges UX research, engineering, marketing, etc.
- › Be careful—personas run on stereotypes



How to make and use personas

- › Base them on empirical data, not assumptions (or else, call them proto-personas)
- › Keep them simple, visual, succinct (1 page)
- › Create them collaboratively, with leadership buy-in
- › Include them in your **scenarios** and **storyboards** and **journey maps**
- › Post them around the office, refer to them in meetings



Scenarios

- › Are predominantly textual, but can be video
- › Describe a user in a situation, often interacting with a current or future system
- › Range from a sentence to a paragraph or more



Example: User Scenario

Alice is 67 years old. She is visiting her children and grandchildren for the holidays. Her flight has a 4-hour layover at an airport she has never visited before. Within the layover time, she would like to shop for her family, grab dinner, and find places with charging ports for her device. Additionally, she prefers to complete most activities near the departure gate so that she can be on time for the flight. She is aware of the presence of visual maps with information on the facilities, shops and directions, but, these maps are not accessible because of her visual impairment. She also likes to ask questions of airport staff, but they are hard to find and are not always understanding or helpful. She wishes that she could access the information provided by the maps through accessible technology she already owns.



Example: Task Scenario

You're planning a vacation to New York City, March 3 – March 14. You need to buy both airfare and hotel. Go to the American Airlines site and jetBlue Airlines site and see who has the best deals.



Why create scenarios?

- › Inexpensive way to imagine user contexts, needs, interaction
- › Flexible use throughout design cycle: establishing requirements, design ideation, development, and evaluation
- › Communication tool amongst product managers, designers, engineers, marketing, etc.



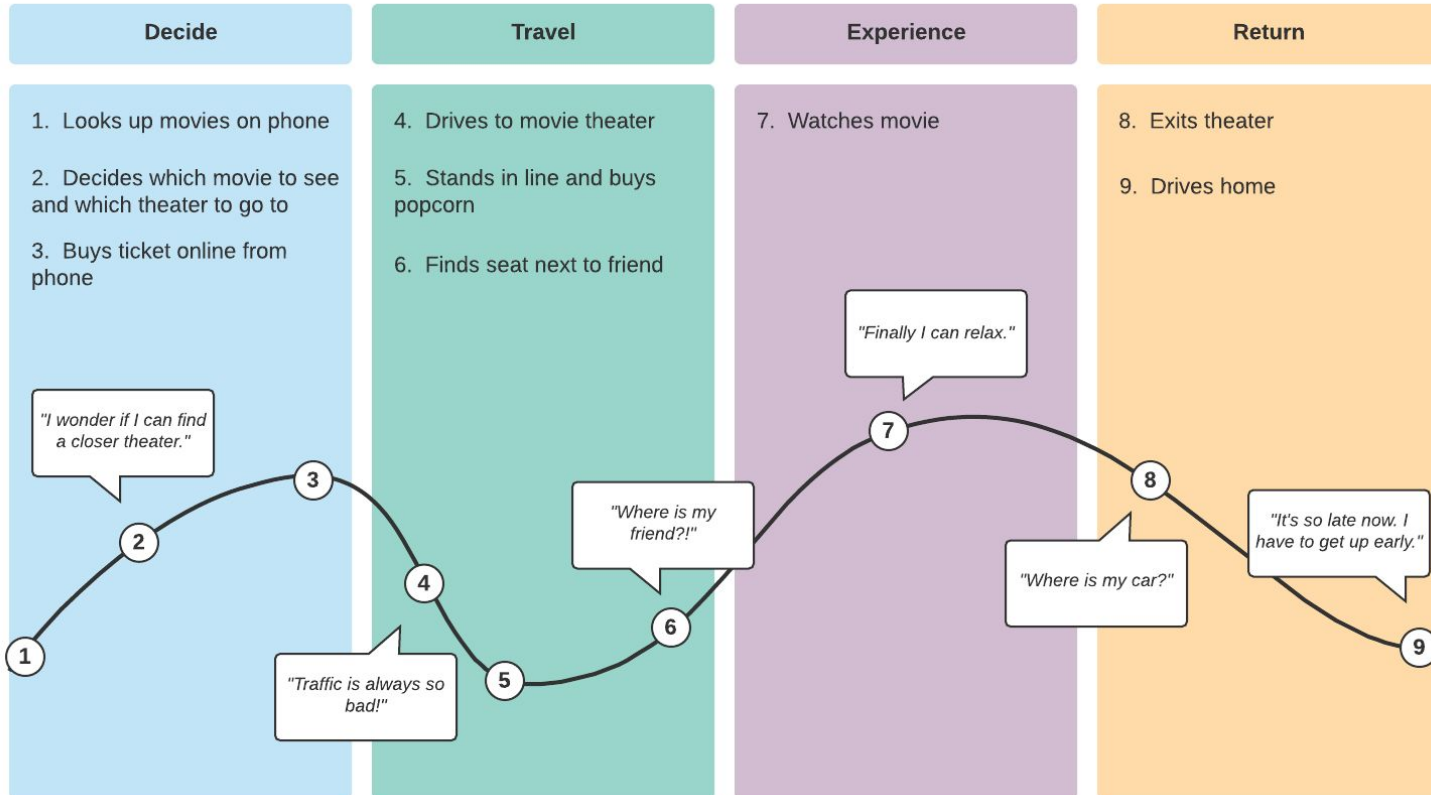
Sarah

Scenario

Sarah is going to the movies. She is excited to go out for the night and will meet her friend at the theater.

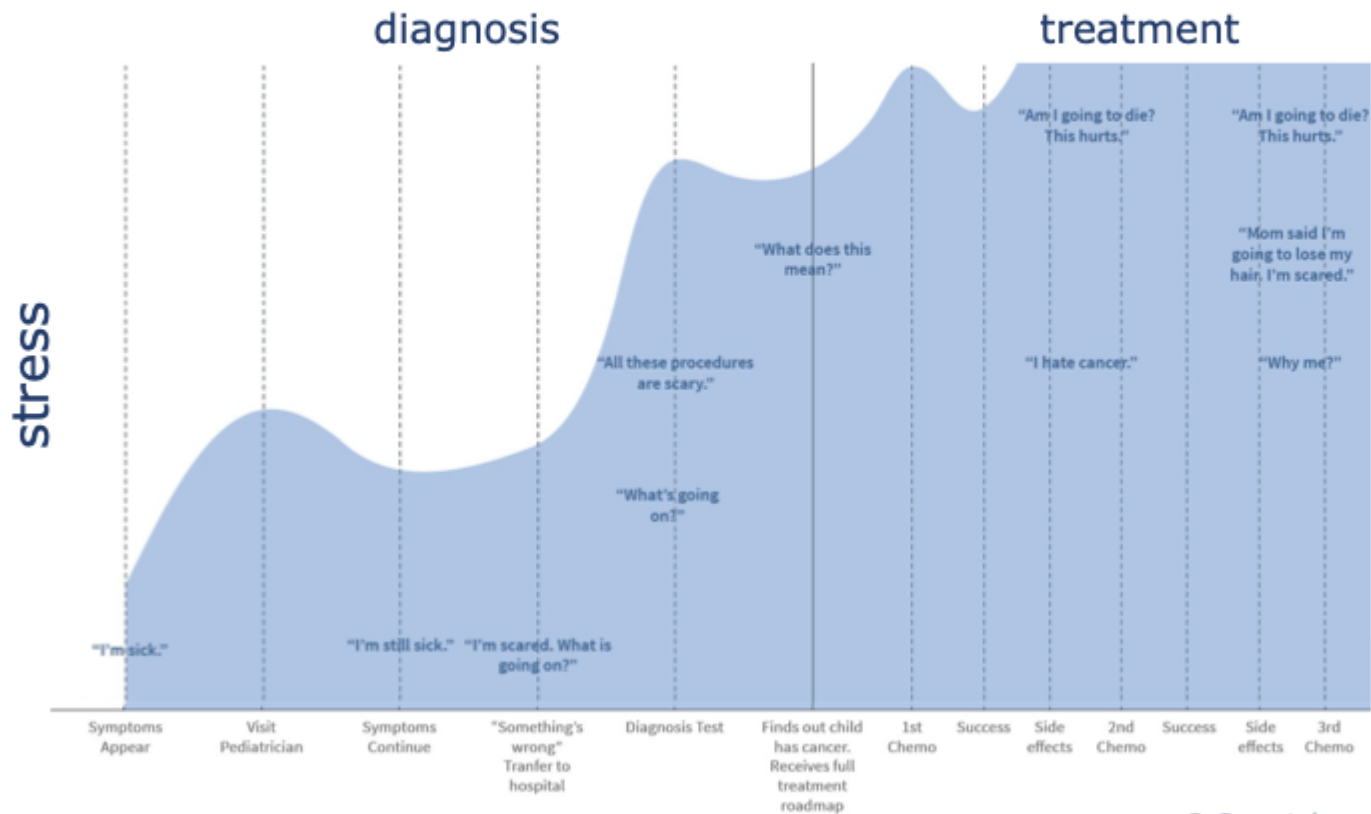
Expectations

- Great movie
- Friendly staff
- Good seat



EXPERIENCE MAP *Example (Pregnancy)*

TRIMESTER	1ST	2ND	3RD
ANXIETY LEVELS + COMMON TESTS	Positive Pregnancy Tests Urine analysis Maternal Serum Chorionic Villus Sampling	Fetal Development and Gender Determination Fetal Heartbeat Monitoring Alpha-fetoprotein screening, hCG, estiol, inhibin Glucose Tolerance Tests	3D Ultrasound Fetal Non-Stress Tests
SHARING	Partner Close Friends/Family	Other Friends/Work	Obvious in Public
PLANNING	Name Generation	Maternity Leave Plans Nursery and Supplies Prep	Birthing Classes Baby Shower + Hospital Bag
PHYSICAL EXPERIENCE Energy Weight Discomfort			



© Sproutel



Journey Maps Generally Include...

- › Sequence of actions / touchpoints → over time
- › Persona
- › Graphs
 - Emotions
 - Physical status (weight)
- › Phases of the journey
- › Journey title
- › Scenario
- › Quotes at different parts of the journey



What's a Journey Map?

- › The use of personas alone may not be enough to make design decisions.
- › Journey maps:
 - Add a **third dimension (time)** to the static nature of personas.
 - Offer high-level overview of a user's **interactions** over time.
 - Illustrate how a user's **emotional** state, needs, and goals change over time.
 - Focus on user **patterns** and the entirety of their experience.



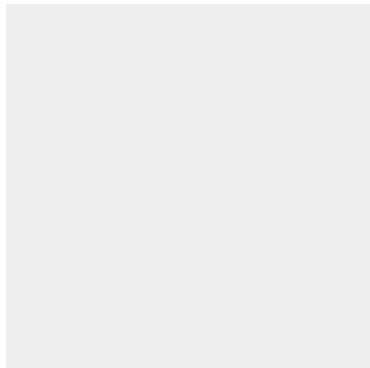
Considerations

- › From whose **perspective** the journey is taking place
- › The **goal** the persona is accomplishing
- › Time sequence of **actions** or tasks performed
- › **Affective state** (thoughts, emotions) at each point of the journey
- › **Tools** or resources they are using
- › Physical **environment**

What Level to Map?

- › used throughout design process at different levels: consumer journeys, user experience, interaction design, etc.
- › Choose the resolution that best represents what you are trying to show
- › Use multiple maps to represent different levels of detail





[Activity 3.2](#)

 40 min

In-Class Activity 2

Create a Persona

- › Form groups of 4 with who researched the same topic.
- › Each briefly shares their chosen IDEO method and key insights from the last exercise.
- › Create a user persona based on your findings and add it to the slide deck.

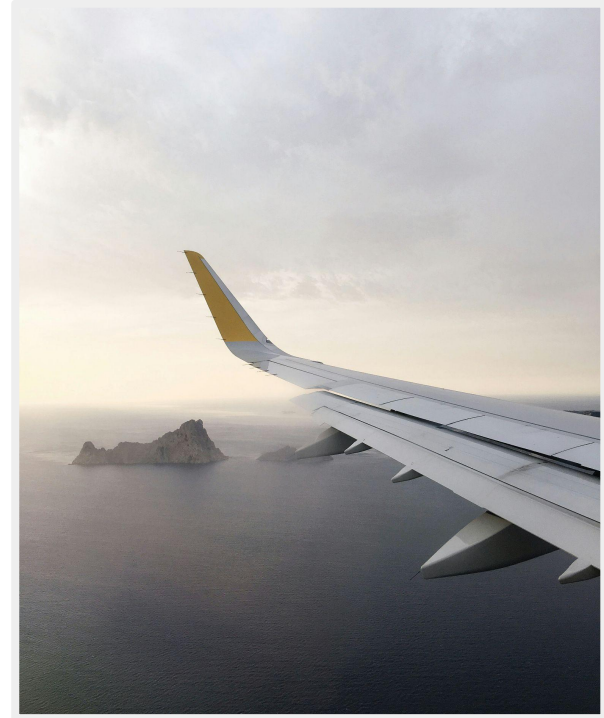
Personas: What Did You Think?

Path

- UX Research (~20min)
- Interviews, Observations, & Contextual Inquiry (~45min)
- Activity 1 + Break (~15min + 15min)
- Personas & Journey Maps + Activity 2 (~15min + 50min)
- Coming Up (~5min)

Assignment: “Persona and Journey Map”

- Create a persona and journey map for an airline app.
- Use personal experience, research, and imagination to define customer needs, expectations, and app touchpoints.
- Submit both as one document by this Sunday.



This Week

- › Complete Week 3 Deliverable **Persona and Journey Map**
- › Do Week 4 **readings**

Have a great week!